

BRIEF COMMUNICATION

Early clinical experience of bacteriophage therapy in 3 lung transplant recipients

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Bacteriophage therapy (BT) uses bacteriophages to treat pathogenic bacteria and is an emerging strategy against multidrug-resistant (MDR) infections. Experience in solid organ transplant is limited. We describe BT in 3 lung transplant recipients (LTR) with life-threatening MDR infections caused by *Pseudomonas aeruginosa* (n = 2) and *Burkholderia dolosa* (n = 1). For each patient, lytic bacteriophages were selected against their bacterial isolates. BT was administered for variable durations under emergency Investigational New Drug applications and with patient informed consent. Safety was assessed using clinical/laboratory parameters and observed clinical improvements described, as appropriate. All patients received concurrent antibiotics. Two ventilator-dependent LTR with large airway complications and refractory MDR *P. aeruginosa* pneumonia received BT. Both responded clinically and were discharged from the hospital off ventilator support. A third patient had recurrent *B. dolosa* infection following transplant. Following BT initiation, consolidative opacities improved and ventilator weaning was begun. However, infection relapsed on BT and the patient died. No BT-related adverse events were identified in the 3 cases. BT was

Abbreviations: BAL, bronchoalveolar lavage; BT, bacteriophage therapy; CF, cystic fibrosis; FDA, Food and Drug Administration; IV, intravenous; LTRs, lung transplant recipients; MDR, multidrug-resistant; PFU, plaque-forming units.

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well tolerated and associated with clinical improvement in LTRs with MDR bacterial infection not responsive to antibiotics alone. BT may be a viable adjunct to antibiotics for patients with MDR infections.

KEYWORDS

antibiotic drug resistance, antibiotic: antibacterial, clinical research/practice, infection and infectious agents – bacterial, infectious disease, lung disease: infectious, lung transplantation/pulmonology, translational research/science

1 | INTRODUCTION

Lytic bacteriophages are viruses that attach to bacterial surface receptors and inject their genomic DNA/RNA into host cytoplasm, triggering bacterial lysis. Most bacteriophages are very specific to 1 or a few related strains/species of bacteria and the large differences between bacterial prokaryotic and human eukaryotic cells prevent cross-infection.¹ Bacteriophage therapy (BT) involves the identification and administration of bacteriophages to target specific pathogenic bacteria. Experience with BT is over a century old but early clinical potential is largely forgotten.² Limited antibiotic options for multidrug-resistant (MDR) bacteria and technical advances in bacteriophage purification have led to renewed interest. Case reports have highlighted the use of BT for MDR organisms including *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Staphylococcus aureus*.^{3–6} Recent case reports describe the successful use of genetically engineered BT in the treatment of disseminated *Mycobacterium abscessus* infection in a lung transplant recipient (LTR)⁷ as well as use of BT to treat MDR *P. aeruginosa* infection in a cystic fibrosis (CF) patient who went on to successful lung transplant.⁸

LTRs have uniquely high rates of MDR infections, representing an attractive population to assess the safety, tolerability, and potential efficacy of adjunctive BT. In addition to the pretransplant prevalence of MDR organisms and the risk of posttransplant recolonization, uncontrolled allograft infection can cause anastomotic and large-airway complications, including dehiscence and stenosis.^{9,10} When treating MDR bacterial infections, there are limited antibiotic options, many of which have adverse events (AE), including nephrotoxicity and marrow suppression,¹¹ which are especially significant in LTRs given the ongoing need for calcineurin and cell cycle inhibitors.¹²

We describe our clinical experience regarding BT to treat MDR infections in 3 LTRs.

2 | METHODS

Three LTRs with lower respiratory tract infections were identified at the University of California San Diego, the Hospital of the University of Pennsylvania, and the Cleveland Clinic as potential candidates for BT. Two were infected with MDR *P. aeruginosa* and 1 with MDR *B. dolosa*.

Phages were obtained from the following sources: AmpliPhi Biosciences Corporation (San Diego, CA), Naval Medical Research Center (Fort Detrick, MD) and Adaptive Phage Therapeutics (Gaithersburg, MD).

2.1 | BT administration

Table 1 describes details of the phages used including concentration, frequency of dosing, and route of administration. Endotoxin concentration refers to the amount of bacterial endotoxin lipopolysaccharide produced during phage mediated lysis of the host bacteria during phage propagation for production. The US Food and Drug Administration (FDA) mandates a maximum administered dose of endotoxin to be <5 endotoxin units/kg body weight per hour.

Intravenous (IV) bacteriophage solutions were prepared just before administration when possible; however, stability testing was performed. We used the Pari eFlow[®] rapid nebulizer for respiratory delivery of the phage solution to Patient 1.

2.2 | Susceptibility testing

Two methods were used to assess bacterial susceptibility to phages. Susceptibility to AB-PA01 and AB-PA01-m1 was tested using a version of the double agar overlay method with nutrient broth-based liquid and agar media.¹³ Isolates were classified as susceptible when individual plaques were observed in drop tests with serial dilutions. Susceptibility testing to the Navy phage cocktails 1 and 2 as well as *B. dolosa* phage was assessed by inoculating bacterial suspensions (10^4 colony-forming units/mL) with bacteriophages individually and in combination in a 96-well microtiter plate containing tryptic soy media in 1% (vol/vol) tetrazolium dye and incubated at 37°C in a Biolog[®] machine for 24 hours. Bacterial respiration reduced the tetrazolium dye, which changed the color of the media to purple. This color change was depicted as relative units of bacterial growth.¹⁴

2.3 | Bacteriophage detection in serum and respiratory samples

Glycerol-amended serum and bronchoalveolar lavage (BAL) samples were thawed, serially diluted, and spotted onto an agar overlay. BAL samples were passed through a 0.2- μ m polyethersulfone syringe

TABLE 1 Details of the bacteriophages used in the case series

Patient	Bacteriophage cocktail	Component bacteriophages	Concentration per unit IV dose	Endotoxin concentration per unit IV dose	Dosing route and frequency
Case 1	AB-PA01 ^a	Pa193, Pa204, Pa222, and Pa223	4×10^9 PFU/mL	0.2 EU/mL	IV every 6 h. Nebulized every 12 h ^b
Case 1	AB-PA01 m1 ^a	Pa193, Pa204, Pa222, Pa223, and Pa176	5×10^9 PFU/mL	8.3 EU/mL	IV every 6 h. Nebulized every 12 h ^b
Case 1	Navy phage cocktail 1 ^c	Paφ1, PaSKWφ17, and PaSKWφ22	1×10^9 PFU/mL	7.3×10^3 EU/mL	IV every 2 h. Nebulized every 6 h ^b
Case 1	Navy phage cocktail 2 ^c	PaATFφ1 and PaATFφ3	5×10^7 PFU/mL	1.5×10^3 EU/mL	IV every 4 h
Case 2	AB-PA01 ^a	Pa193, Pa204, Pa222, and Pa223	4×10^9 PFU/mL	22.0 EU/mL	IV every 12 h
Case 3	Single lytic phage	BdPF16phi4281 ^d	5.3×10^6 PFU/mL – 3.5×10^7 PFU/mL	200 EU/mL	IV once daily for the first 2 wks and then every 12 h for 4 wks.

Abbreviations: EU, endotoxin units; IV, intravenous; PFU, plaque-forming units.

^aAB-PA01 is an investigational product, formulated to contain equal concentrations of 4 *Pseudomonas aeruginosa* phages targeting ≈80% of clinical *P. aeruginosa* isolates, available as 1-mL sterile solution (NCT03395743). AB-PA01 was produced at AmpliPhi Good Manufacturing Practices (GMP) facility using a proprietary production process consisting of upstream fermentation and several downstream purification steps. AB-PA01 m1 was produced using the AB-PA01 component phages manufactured in the same way, with the addition of Pa176 produced in a non-GMP facility using analogous methods.

^bNebulized dose was 4 times the IV dose.

^cNavy Phage cocktails 1 and 2 consisted of bacteriophages isolated from wastewater collected from water treatment facilities in Germantown, Frederick, or Laurel, Maryland using common soft agar overlay techniques.²⁷

^dBdPF16phi4281 was produced by Adaptive Phage Therapeutics and was amplified on the patient bacterial strain on solid growth media. Extracted phage lysate was clarified by centrifugation and filtration. The phage was further purified via isopycnic centrifugation on a cesium chloride gradient followed by multiple rounds of dialysis in an appropriate buffer solution to remove residual cesium. The purified phage was diluted with enough excipient to reduce the endotoxin concentration to below the maximal permissible hourly limit of EU/kg (where kg=body weight in kilograms). The final formulated phage was filtered through a 0.22-μm filter, aseptically filled into single-use vials, and stored at 4°C until use.

^ePhages were prepared and supplied in 5 different batches over the course of treatment.

filter prior to dilution. Positive controls were prepared by spiking an aliquot of bacteriophage-free serum/BAL sample from the same patient with a known quantity PFU/mL of each AB-PA01 component phage.

2.4 | Ethics/regulatory approval

Administration of BT was conducted at each institution with individual emergency Investigational New Drug applications from the FDA, Institutional Review Board notification, Biohazard Use Authorization, and patient informed consent.

2.5 | Safety

Safety was assessed using clinical and laboratory parameters, including AE log, vital sign monitoring, and serial complete blood count and comprehensive metabolic panels. All patients were treated in the inpatient setting and were followed closely to assess clinical resolution and/or improvement of infection (ie,

resolution of fever, leukocytosis, sputum production) as well as development of AE.

3 | RESULTS

3.1 | Clinical courses

3.1.1 | Patient 1

Patient 1 was a 67-year-old man who underwent bilateral lung transplant for hypersensitivity pneumonitis. His posttransplant course was complicated by primary graft dysfunction, prolonged mechanical ventilation, right main stem bronchial stenosis, and multiple episodes of *P. aeruginosa* pneumonia with antibiotic susceptibilities noted in Table 2. He required serial bronchial dilation and stenting, developed chronic lung allograft dysfunction requiring extracorporeal photopheresis, and progressive kidney injury necessitating hemodialysis. He had 2 distinct episodes of MDR *P. aeruginosa* pneumonia that were treated with BT along

TABLE 2 Antibiotic susceptibility of baseline clinical bacterial isolates as determined according to Clinical & Laboratory Standards Institute guidelines²⁸

Organism (patient no.)	Susceptible	Intermediate	Resistant
<i>Pseudomonas aeruginosa</i> (Patient 1, Episode 1)	Amikacin, tobramycin	Gentamicin, colistin	Aztreonam, piperacillin-tazobactam, meropenem, ceftazidime, cefepime, ciprofloxacin, ceftazidime-avibactam
<i>Pseudomonas aeruginosa</i> (Patient 1, Episode 2)	Amikacin, tobramycin, piperacillin-tazobactam	Ceftazidime, colistin	Aztreonam, gentamicin, meropenem, doripenem, cefepime, ciprofloxacin
<i>Pseudomonas aeruginosa</i> (Patient 1, Episode 2)	Tobramycin, colistin	-	Aztreonam, amikacin, gentamicin, piperacillin-tazobactam, meropenem, doripenem, ceftazidime, cefepime, ciprofloxacin
<i>Pseudomonas aeruginosa</i> (Patient 2)	Colistin	Piperacillin-tazobactam	Amikacin, cefepime, ceftazidime, ceftazidime-avibactam, gentamicin, levofloxacin, meropenem, tobramycin
<i>Burkholderia dolosa</i> (Patient 3)	Minocycline	-	Ceftazidime, ceftazidime/avibactam, ceftolozane/avibactam, levofloxacin, meropenem, ticarcillin/clavulanate, trimethoprim/sulfamethoxazole

with concomitant antibiotics and then was maintained on BT as suppression.

Episode 1

Patient 1 received a 2-week course of IV and nebulized AB-PA01, a prefixed phage cocktail consisting of 4 *P. aeruginosa* phages with a staggered start of the latter (to assess differential bacteriophage BAL concentrations based on mode of delivery) as an adjunct to systemic antibiotics (piperacillin-tazobactam and colistin). At baseline, the patient required supplemental oxygen through tracheostomy at 10 L/min, 35% fraction of inspired oxygen and had copious purulent respiratory secretions. By 2 weeks of BT, he had significantly decreased inflammation and minimal respiratory secretions noted on bronchoscopy. The patient started to ambulate by day 16, and talked via a Passy-Muir valve for several hours on day 18. Concomitant antibiotics were stopped on day 18. Nebulized AB-PA01 was extended by an additional week without systemic antibiotics in an attempt to repopulate the airways with normal respiratory flora. By day 21, BAL cultures included non-*P. aeruginosa* bacterial species, suggesting re-establishment of respiratory flora (data not shown). The patient completed inhaled AB-PA01 therapy on day 29.

Episode 2

On day 46, Patient 1 clinically decompensated following a surveillance bronchoscopy with development of fever, leukocytosis, respiratory failure requiring mechanical ventilation, and septic shock requiring vasopressors. Respiratory cultures grew mucoid MDR *P. aeruginosa*. Systemic antibiotics (piperacillin-tazobactam, tobramycin, and inhaled colistin) were restarted and BT was used. BT consisted of distinct courses of AB-PA01-m1 (prefixed combination AB-PA01 + 1 new specific phage) and Navy phage cocktail 1 (personalized phage cocktail), with clinical resolution of pneumonia.

Suppression phase

The patient received suppressive BT (Navy phage cocktails 1 and 2, personalized phage combination) from days 93-150. During this period as well as the following 3 months, there was no active

P. aeruginosa pneumonia. The patient's tracheostomy was de-cannulated, he made progress with physical rehabilitation, and was discharged from the hospital. No AE related to BT occurred during treatment.

3.1.2 | Patient 2

Patient 2 was a 57-year-old woman with non-CF bronchiectasis who was colonized with a MDR *P. aeruginosa* sensitive to only colistin (Table 2). She was bridged to transplant with ambulatory venovenous extracorporeal membrane oxygenation. Posttransplant, she had significant bilateral airway ischemic injury and developed recurrent MDR *P. aeruginosa* infections. She underwent tracheostomy to facilitate ventilator weaning but severe airway stenosis limited her progress despite serial balloon dilation. She also developed *Mycobacterium abscessus* pulmonary infection, initially treated with imipenem, tigecycline, and inhaled amikacin. As a result of nephrotoxic antibiotic exposure, she had progressive renal failure and required hemodialysis. Due to an inability to clear *P. aeruginosa* from respiratory cultures and concern that ongoing infection was preventing airway healing, BT was pursued.

She was treated with a 4-week course of IV AB-PA01 and continued only on inhaled colistin concomitantly. There were no BT-related AE and the patient clinically responded to treatment. Markers of inflammation (serum sedimentation rate, C-reactive protein, and procalcitonin) did not change significantly during AB-PA01 administration.

No additional *P. aeruginosa* was cultured from BAL specimens following the start of BT treatment until 60 days after completion of AB-PA01 treatment. The isolate grown at day 60 and subsequent strains showed improved antibiotic susceptibility. Additional infections were successfully treated with piperacillin-tazobactam. She was successfully weaned from the ventilator by day 90 and was discharged from the hospital. She did not grow additional *M. abscessus* in the 7 months following BT administration and mycobacterial therapy was stopped after 90 days of treatment. She was subsequently readmitted with an aspiration pneumonia but did not grow MDR *P. aeruginosa*. She remains on inhaled colistin for suppressive therapy.

3.1.3 | Patient 3

Patient 3 was a 28-year-old woman with CF colonized with MDR *B. dolosa* sensitive only to minocycline (Table 2). At the time of bilateral lung transplant, she received perioperative systemic trimethoprim/sulfamethoxazole, minocycline, and meropenem with amikacin irrigation intraoperatively. Four days following transplant, *B. dolosa* was grown from bilateral pleural spaces and 8 days posttransplant grew *B. dolosa* from an endobronchial culture. The pleural spaces remained infected despite systemic and intrapleural antimicrobial therapies and effective drainage. Two months after transplant an isolate was sent for identification of a lytic bacteriophage, but a suitable candidate was not identified until 3 months later. In the interim she developed recurrent episodes of pneumonia due to *B. dolosa*. She was treated with minocycline, meropenem, and trimethoprim/sulfamethoxazole (based on in vitro susceptibilities) but experienced ongoing drug-related toxicities and remained ventilator-dependent.

She eventually responded favorably to a 6-week course of combined therapy with extended infusions of ceftazidime-avibactam and piperacillin-tazobactam. However, pneumonia recurred 3 weeks after completing that course. Now 6 months posttransplant, the same antibiotic regimen was resumed and, 1 week later, BdPF16phi4281 (a single lytic phage with in vitro anti-*B. dolosa* activity) was added (Table 1). She had no BT-related AE. After BT initiation, she had improvement of her fever and leukocytosis, decreased airway secretions, improved consolidations on imaging, and decreased *B. dolosa* burden on BAL culture. After 6 weeks of combined therapies she was ambulating 660 feet and became ventilator free for up to 12 consecutive hours. However, she developed progressive renal failure ascribed to calcineurin inhibitor toxicity, and, 2 weeks later, had severe hemorrhage from a splenic artery pseudoaneurysm requiring splenectomy. Postoperatively, she developed progressive liver injury thought to be multifactorial (including existing underlying CF hepatopathy, hypotension, and drug-induced injury). On week 10 of BT, she developed septicemia from *B. dolosa* bloodstream infection and pneumonia. Based in part on a clinical case report and updated susceptibilities, her antibiotics were changed to extended infusion meropenem, extended infusion ceftazidime-avibactam, minocycline, and inhaled tobramycin.¹⁵ Her sepsis resolved, bloodstream infection cleared, and her respiratory status improved. However, she developed progressive liver failure with concern for drug-induced toxicity, including from minocycline, dapson, and posaconazole prophylaxis. She had completed the planned 12-week course of BT at that time and it was decided that the treatment was unlikely to be of ongoing benefit. Antibiotics were also stopped due to toxicities. A liver biopsy demonstrated severe cholestasis and hepatocyte swelling. After 1 week off antibiotics, she again developed pneumonia and demonstrated heavy growth of *B. dolosa* from a BAL sample. Despite antibiotic treatment, she clinically worsened, and transitioned to hospice. She died 11 months following lung transplantation. No autopsy was obtained.

3.2 | Bacteriophage detection in serum and respiratory samples (Patient 1)

No bacteriophages were detected prior to BT initiation. Once administration was started, no viable AB-PA01 or AB-PA01-m1 phages were detected in serum samples drawn within 30 minutes of the next IV dose. High recovery of spiked phages from serum samples suggested that serum itself did not interfere with detection of phages. Each IV dose of AB-PA01 consisted of 4×10^9 plaque-forming units (PFU). Approximately 4×10^7 PFU/mL were recovered in BAL samples obtained 3 days after initiation of IV AB-PA01 therapy (and prior to administration of inhaled AB-PA01) and day 29 (when the patient had received inhaled AB-PA01 only for the previous week) (Figure 1). Thus, IV and inhaled AB-PA01 each reached similar pulmonary concentrations, with both modalities exhibiting intrapulmonary phage recovery.

3.3 | In vitro bacterial susceptibility to individual phages and phage cocktails

3.3.1 | Patient 1

P. aeruginosa isolates were susceptible to all 4 component bacteriophages of AB-PA01 while on BT (Table 3). Day 36 isolate, cultured 7 days after cessation of AB-PA01, was resistant to all 4 component bacteriophages of AB-PA01. Pa176, which was added to create AB-PA01-m1, was chosen to target this isolate. All subsequent *P. aeruginosa* isolates were susceptible to at least 1 of the AB-PA01-m1 components.

Figure 2 shows bacterial time-kill curves of Patient 1's clinical isolates by *P. aeruginosa* Navy phage cocktails 1 and 2 (individual components as well as both cocktails). Resistance developed over time to Navy phage cocktail 1 and its individual components as indicated by a shift in the time-kill curves to the left in Panel B versus Panel A. Panel B also shows lytic activity of Navy phage cocktail 2 compared to 1.

3.3.2 | Patient 2

No *P. aeruginosa* isolates grew for the patient until 2 months after BT was completed. Two pre-BT isolates were susceptible to AB-PA01 specifically to 2 of the 4 component phages (not shown).

3.3.3 | Patient 3

B. dolosa isolates were not tested for phage susceptibility during or after therapy. Two baseline isolates prior to BT were susceptible (not shown).

3.3.4 | Adverse events

IV and nebulized formulations of bacteriophages used in the series were well tolerated without AE attributed to bacteriophage administration.

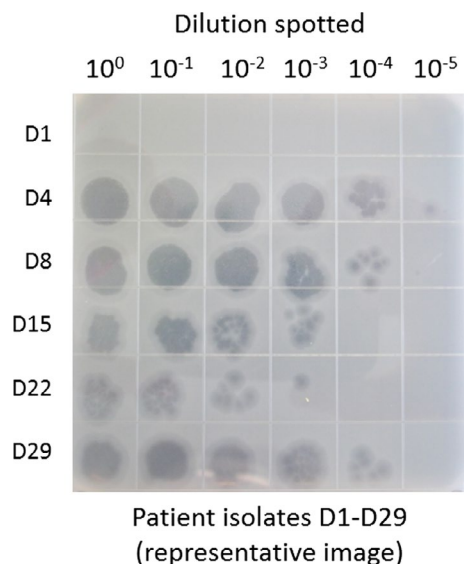


FIGURE 1 Detection of active bacteriophages in respiratory samples during use of AB-PA01 for Patient 1. Collected BAL samples were serially diluted in phage buffer and the phage titer of AB-PA01 was assessed in triplicate using the double agar overlay method with nutrient broth-based liquid and agar media. Bacteriophage was absent in BAL before BT (day 1) and was measured at high concentrations during therapy (days 4–29). BAL, bronchoalveolar lavage; BT, bacteriophage therapy [Color figure can be viewed at wileyonlinelibrary.com]

4 | DISCUSSION

In this case series of LTRs with MDR bacterial infections, BT was well tolerated with no clear BT-related adverse events. When used in combination with systemic or inhaled antibiotics, we noted clinical improvement. In Patient 2, we observed microbiological eradication, which had not been achieved at any prior time point with antibiotics alone. This case series suggests that bacteriophages are a novel adjunct antibacterial therapy for the treatment of MDR bacterial infections among LTRs.

Recent clinical use of BT has covered a range of applications, including skin and soft tissue infections, bacteremia, sepsis, osteomyelitis, otitis media, and indwelling medical devices.^{3,4,6,16–18} There has been growing interest in BT for patients with suppurative advanced lung disease, and animal models have shown BT efficacy for MDR organisms found in CF such as *Burkholderia cenocepacia* and *Klebsiella pneumoniae*.^{19–21} BT may be useful in LTRs for several reasons: (1) to kill specific pathogens while theoretically allowing nonpathogenic microorganisms to repopulate the pulmonary microbiome, (2) to apply directional selection that might render the bacteria more susceptible to antibiotics or less fit in vivo even at the cost of developing phage-resistant bacteria,^{22,23} and (3) to penetrate and lyse biofilm-state bacteria that are traditionally less susceptible to antibiotics, such as on stents. Our cases suggest that a combination of the above factors in addition to systemic antibiotics may act together to resolve infection.^{24,25} In the 2 LTRs with MDR *P. aeruginosa* infections, adjunctive BT may have contributed to clinical improvement through a combination of these

mechanisms. Other factors affecting clinical resolution of illness may include phage penetration into the infected site (both *P. aeruginosa* cases were pneumonias while the *B. dolosa* pneumonia extended into the pleural cavity), duration of the infection prior to BT treatment, and/or development of phage-mediated modulation of immune response directed towards the pathogen.

Development of resistance to the bacteriophage is a concern when utilizing BT.³ In Patient 1, one isolate resistant to AB-PA01 appeared after cessation of AB-PA01 and 1 isolate resistant to AB-PA01-m1 was noted following cessation of AB-PA01 m1, but none while on BT. *P. aeruginosa* isolates also developed resistance while on Navy phage cocktail 1 and this was overcome with the use of a personalized approach and identification of new phages active against the *P. aeruginosa* isolates. In terms of the *B. dolosa* case, susceptibility testing of the isolates obtained during BT was not done. We hypothesize that the low concentration titer of this phage (10^{6-7} PFU/mL), less frequent dosing interval, and therapy with a single phage rather than combination, may have led to either inadequate concentration at the site of infection and/or development of resistance (though we did not test this specifically).

There were no significant BT-related AEs in the 3 patients. Although Patient 3 did not survive, the sequence of events that led to multiorgan failure are more plausibly ascribed to postsplenectomy complications and liver disease rather than BT. Patient 2 did not have clinical evidence of acute cellular or antibody-mediated rejection following BT.

Although Patient 1 subsequently grew *P. aeruginosa* strains in respiratory specimens, the antibiotic resistance patterns differed from the original isolates (data not shown). This was true for Patient 2 as well, whose *P. aeruginosa* isolates 2 months after BT cessation had different antimicrobial susceptibility patterns than the original isolates and demonstrated increased in vitro susceptibility to several antibiotic classes.

In this case series, we described 2 different approaches to the clinical use of BT in a real-world setting. One was the use of a pre-existing investigational product (Patients 1 and 2) and the other was a personalized approach, including the isolation and development of specific bacteriophage(s) against the patient's specific clinical isolates (Patients 1 and 3). We believe that both approaches may be clinically relevant, depending on the infection being treated. For bacteria that lack susceptibility to pre-existing products, personalization may offer the only feasible option of BT.

In addition to the potential of BT to treat MDR bacterial infections, other potential benefits include avoidance of antibiotic toxicity and potential changes in microbial susceptibility patterns rendering the organism more susceptible to antibiotics.²⁶

Challenges in obtaining BT for our patients were based mainly on the experimental nature of this form of therapy. Even with the use of a pre-existing phage combination, such as AB-PA01, there was a delay of several weeks from sending patients' bacterial isolates to a research laboratory for susceptibility testing, development of a treatment plan, approval from the FDA, and then the

TABLE 3 Susceptibility of serial *Pseudomonas aeruginosa* to AB-PA01, AB-PA01-m1, and their component phages (Patient 1)^a

Treatment				Timepoint	Susceptibility to Product and Component Phages	
					AB-PA01 ^b	AB-PA01-m1 ^c
				14-d pretreatment	Susceptible (all 4 phages)	NA
				9-d pretreatment	Susceptible (all 4 phages)	NA
AB-PA01	Neb	IV	D 1 ^d	Susceptible (all 4 phages)	NA	
			D 4	Susceptible (all 4 phages)	NA	
			D 8	Susceptible (all 4 phages)	NA	
			D 15	Susceptible (all 4 phages)	NA	
			D 22	Susceptible (Pa222, Pa223)	NA	
			D 29	Susceptible (Pa193, Pa204)	NA	
			3-wk interregnum		D 36	Resistant ^e
		D 46	Susceptible (all 4 phages)	Susceptible (all 5 phages)		
AB-PA01-m1	Neb	IV	D 53	(no culture)	(no culture)	
			D 56	NA	Susceptible (Pa193, Pa204, Pa176)	
			D 60	NA	Susceptible (Pa193, Pa204, Pa176)	
Navy 1	Neb	IV	D 64	NA	Susceptible (Pa193, Pa204, Pa176)	
			D 69	NA	Susceptible (Pa176)	
			D 74 ^f	NA	Susceptible (Pa176)	
			D 77	NA	No	
AB-PA01-m1	Neb	IV	D 78	(no PA)	(no PA)	
			D 82	(no PA)	(no PA)	
			D 88	(no PA)	(no PA)	
			D 92	(no PA)	(no PA)	
Navy 1		IV	D 110	NA	Susceptible (Pa222, Pa223)	
Navy 2			D 114	NA	Susceptible (all 5 phages)	
			D 141 ^f	NA	Susceptible (Pa222, Pa223)	

NA, not applicable, since the product was neither in use nor being considered for use at this time point.

^aSamples were tested in triplicate using independent bacterial cultures and serial dilutions of standardized bacteriophage preparations. Isolates were only classified as "susceptible" when individual plaques were observed in drop tests with serial dilutions. Susceptibility to individual phages is indicated in parentheses.

^bAB-PA01 contains 4 phages.

^cAB-PA01-m1 is AB-PA01 plus 1 additional phage (total 5 phages).

^dSamples was collected prior to first bacteriophage therapy.

^eThis isolate, collected during the period when no phage product was being administered, prompted the addition of Pa176 to AB-PA01, leading to AB-PA01-m1.

^fTwo isolates were obtained from these samples, both of which yielded similar phage susceptibility patterns.

actual logistics of shipping to the hospital (though this timeline has shortened with subsequent cases, as all parties have become more accustomed to the process). The personalized phage approach took longer, particularly for the third case, because a "phage hunt" needed to be carried out to find lytic phages for a particular bacterial isolate, genotypic characterization, growth in sufficient quantities, and endotoxin removal so that it would be safe to use in a patient. With the growth of extensive phage libraries both in the academic and commercial setting, we hope that the time to match a lytic phage to a bacterial pathogen and subsequent production of suitable safe products will be reduced.

We report early clinical experience of BT in LTRs targeting drug-resistant pathogens that are common in this patient population. Many questions not directly addressed in this study require further investigation. These include optimizing the timing, route, and frequency of administration (inhaled/intravenous/both), and duration of therapy; monitoring for the development of phage resistance, impact of dialysis and renal and/or hepatic insufficiency on phage kinetics; bacteriophage-specific antibody responses and complement-mediated clearance, interactions with transplant medications including antiviral therapy; surveillance for bacteriophage and allograft directed humoral immune response, and use of BT as suppressive versus acute treatment.

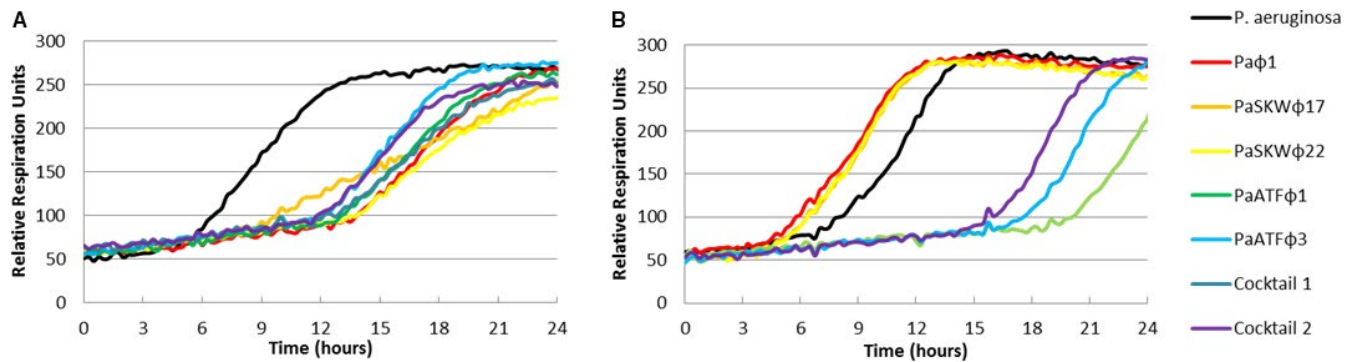


FIGURE 2 Effects of Navy phage cocktails 1 and 2 on the *Pseudomonas aeruginosa* isolates of Patient 1. *P. aeruginosa* isolates (10^4 colony-forming units/mL) were inoculated with bacteriophage cocktail individually and in combination in a 96-well microtiter plate containing tryptic soy media in 1% (vol/vol) tetrazolium dye and incubated at 37°C in a Biolog[®] machine for 24 hours to evaluate bacteriophage killing activities. Bacterial respiration reduced the tetrazolium dye, which changed the color of the media to purple. This change of color was recorded by a camera and depicted as relative units of bacterial growth. A, The graphs depict the growth of *P. aeruginosa* day 29 isolate in the presence of individual bacteriophages (Paφ1, PaSKWφ17, PaSKWφ22, PaATFφ1, and PaATFφ3) along with Navy phage cocktails 1 and 2. B, The graphs depict the growth of *P. aeruginosa* isolate on Day 95 in the presence of individual Navy phages (Paφ1, PaSKWφ17, PaSKWφ22, PaATFφ1, and PaATFφ3) along with Navy phage cocktails 1 and 2

5 | CONCLUSIONS

BT was well tolerated and associated with clinical improvement when used as an adjunct to antibiotics in LTRs with MDR respiratory infections otherwise not responsive to antibiotics alone. BT deserves further evaluation in well-designed clinical trials in the era of increasing antimicrobial resistance.

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DISCLOSURES

The authors of this manuscript have conflicts of interest to disclose as described by the *American Journal of Transplantation*. SL, SM, CLLF, and FR are employed by AmpliPhi Biosciences. MJB is the Chief Medical Officer of Adaptive Phage Therapeutics and owns equity in the company. JRF and BMS are employed by Adaptive Phage Therapeutics.

SAS is a stockholder in Adaptive Phage Therapeutics. RTS is a member of the AmpliPhi Scientific Advisory Board (without compensation).

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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